

Docker Compose and MySQL Integration Documentation

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Environment Setup

- **Project Directory:** C:\Users\ADMIN\IdeaProjects\car-e-pool-api
- **Docker Compose File:** docker-compose.yml
- **Database Image:** mysql:8.0

Initial Run

```
docker-compose up mysql -d
```

- Warning: version attribute in docker-compose.yml is obsolete.
- Actions:
 - Pulled MySQL 8.0 image.
 - Created network car-e-pool-api_default.
 - Created volume car-e-pool-api_carpool-mysql-data.
 - Started container car-e-pool-mysql.

Container Status

```
docker-compose ps
```

- Container: car-e-pool-mysql
- Image: mysql:8.0
- Ports: 0.0.0.0:3308->3306/tcp
- Status: Healthy

Stopping and Removing Containers

Stop

```
docker-compose stop
```

- Error: Invalid interpolation format for SPRING_DATASOURCE_URL.
- Cause: \$ variables need escaping (\$\$).
- Successful stop after correction.

Down with Volumes

```
docker-compose down -v
```

- Removed container car-e-pool-mysql.
- Removed network car-e-pool-api_default.
- Removed volume car-e-pool-api_carpool-mysql-data.

Restarting MySQL Service

```
docker-compose up mysql -d
```

- Successfully recreated network, volume, and container.

```
docker-compose ps
```

- Container running healthy on port 3308.

Service Dependency Issues

- Error: service "carpool-api" depends on undefined service "mysql".
- Cause: Service name mismatch in docker-compose.yml.
- Resolution: Ensure consistent service naming (mysql vs mysql_db).

Flyway Migration Attempt

```
flyway repair
```

- Error: 'flyway' is not recognized as an internal or external command.
- Resolution: Install Flyway CLI or configure Flyway in Maven/Gradle.

API Test Attempt

```
curl --location --request POST 'http://localhost:8080/api/v1/auth/test-login?telegramId=123456&name=Juan'
```

- Error: Port number rejected.
- Cause: Incorrect quoting in Windows CMD.
- Resolution: Use proper quotes or PowerShell syntax.

Lessons Learned

- Remove obsolete version attribute from docker-compose.yml.
- Escape \$ variables in environment definitions with \$\$.
- Ensure service names match across dependencies.
- Install Flyway CLI for database migrations.

- Use correct quoting for curl commands in Windows CMD.

Next Steps

- Fix `SPRING_DATASOURCE_URL` interpolation in `docker-compose.yml`.
- Align service names (`mysql` vs `mysql_db`).
- Install and configure Flyway for schema management.
- Re-test API endpoints with proper syntax.
- Integrate Docker Compose with full Carpool API stack (web + service + repository + domain + database).